

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 1 of 7

(1) Research Program/Project/Study Title: Development of Fruit Cake Infused with Miracle Fruit Wine	
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Program or Project has component (if no proceed to 2)? ☐ Yes ☒ No

Component Title:	None
Component Leader:	None
Academic Rank:	None

(2) Category (please check) ☐ Social Science/Education /Humanities & other related fields ☐ Natural/BiologicalSciences ☒ Food Research/ ☐ Developmental/ Computer/Engineering/ Architecture ☐ Extension ☐ Poster		(3) Research Center [Check the concerned center] ☐ SSITRDC Social Science and Industrial Technology Research Development Center ☐ LRDC Livestock Research Development Center ☒ FFRDC Fisheries and Food Research Development Center ☐ CSRDC Crop Science Research Development Center							
(4) Site of implementation									
		Municipality/City:	Roxas City						
		Province:	Capiz						
		Region:	VI						
(5) Classification <table> <tr> <td>Research:</td> <td>Development:</td> </tr> <tr> <td>☐ Basic</td> <td>☐ Pilot Testing</td> </tr> <tr> <td>☒ Applied</td> <td>☐ Tech. Promotion/Commercialization</td> </tr> </table>		Research:	Development:	☐ Basic	☐ Pilot Testing	☒ Applied	☐ Tech. Promotion/Commercialization	(6) Mode of Implementation ☐ Single Authorship/Proponent ☒ Collaborative Research (within campus) ☐ Collaborative Research (intercampus) ☐ Collaborative Research (external agency)	
Research:	Development:								
☐ Basic	☐ Pilot Testing								
☒ Applied	☐ Tech. Promotion/Commercialization								

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015		Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH		Revision No.	00
				Effective Date	
				Page	Page 2 of 7

(7) Sector/Commodity

Agriculture/Fruit and Vegetables

(8) Discipline/Field of Specialization

Food Development

(9A) Abstract (This has a maximum of 250 words. It contains 1-2 sentences of rationale, followed by objectives, brief methodology, major findings, and implications of impact of the study.)

The typical fruit cake is using red wine for its unique flavor and taste. However, the antioxidants in red wine called polyphenols may help protect the lining of blood vessels in the heart. A polyphenol called resveratrol is one substance in red wine that's received attention for its health benefits.

Further, the fruit of the calabash tree is used for various ornamental purposes. The seeds of the fruit are flat, small, and embedded in the pulp (<https://www.fruitsinfo.com/calabash-tree-tropical-fruits.php>). According to the Philippine Council for Health Research and Development (PCHRD, the fruit and leaf extracts contain flavonoids such as quercetin and anthraquinone, the important phytochemicals for antiangiogenic activities, a process that inhibits the growth and development of new blood vessels in the body." In line with this claim, the realization of miracle fruit wine was created and used as a red wine substitute in making fruit cake.

The general objective was to develop a Fruit Cake infused with miracle fruit wine. Specifically, it sought to: 1. determine the acceptability of the product in terms of four sensory qualities among three treatments; 2. determine if there is a significant difference in the acceptability of the product among three treatments considering the four sensory qualities.

The study used a Completely Randomized design (CRD). It was conducted in three replicates and (1) final process for consumer preference evaluation by the thirty (30) consumers. Scorecards with Nine (9) Point Hedonic Scale was used to obtain the data. The mean and Analysis of variance (ANOVA) were used to analyze the data into an alpha level set at 0.01.

Findings revealed the acceptability of miracle fruit cake in terms of appearance, aroma, taste, and texture, Treatment C was "Like Extremely". Followed by Treatment B, and Treatment C got the lowest mean scores among the three treatments. This means that the more miracle fruit wine infused in fruit cake was preferred by the consumers.

Furthermore, there was a significant difference in the acceptability of the product in terms of four sensory qualities in favor of Treatment C. This implies that Treatment C is the most recommended proportion in miracle fruit wine infused in fruit cake.

(9B) Keywords (Write 3-6 words you used in the study, preferably not found in the title because words in the title are already given keywords. This would also include process, or methodology employed.)

Keywords: Nine (9) Point Hedonic Scale, Sensory Qualities, Wine and Acceptability

(10) Introduction (This includes the rationale, background of the study, and objectives which are not divided by subheadings. Incorporate at least three (3) reviews)

People would choose an artificially made food product that has a less healthy nutritional content over the conventional foods that are said to be "good for the health". Fruits are an important part of a healthy diet, and variety is as important as a quantity that contains essential nutrients that are important for health, growth, and development.

The calabash tree is a small evergreen tropical tree that can grow up to a height of 25 feet and produce fruits up to 25 cm in diameter. It has simple leaves and the bark of the tree is rough. The fruit is large, hard, and green in color and takes about six months to ripen. They cannot be eaten but can be used for various ornamental purposes. The fruits are pollinated by bats and they develop from the trunk of the tree. The seeds of the fruit are flat, small, and embedded in the pulp (<https://www.fruitsinfo.com/calabash-tree-tropical-fruits.php>).

According to the Philippine Council for Health Research and Development (PCHRD), a line agency of the Department of Science and Technology (DOST), the study showed that the calabash tree's "fruit and leaf extracts contain flavonoids such as quercetin and anthraquinone, the important

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 3 of 7

phytochemicals for antiangiogenic activities, a process that inhibits the growth and development of new blood vessels in the body.”

Antiangiogenesis controls the spread of tumor cells in the body by disabling the transport of nutrients toward the cancerous cells, the PCHRD explained. Tumor cells, as they normally do, start with a single cell, which then divides into more cells. “The growth of malignant cells will depend on the availability of specific nutrients being transported by blood vessels”. (<https://businessmirror.com.ph/2015/10/15/calabash-the-miracle-fruit/>).

The typical fruit cake is using red wine for its unique flavor and taste. However, the antioxidants in red wine called polyphenols may help protect the lining of blood vessels in the heart. A polyphenol called resveratrol is one substance in red wine that's received attention for its health benefits.

Thus, the utilization of miracle fruit in wine in making fruit cake was come to realize since it was not yet introduced in the market despite the fact that it has a potential for product and livelihood production of our country. Thus, the study was conducted.

The general objective was to develop Miracle Fruit Cake. Specifically, it sought to: 1. determine the sensory qualities of the product in terms of appearance, aroma, taste, texture among three treatments; 2. find out if there is a significant difference in sensory qualities of the product among three treatments.

The present study were based on the following prior arts such as:

CN109864101A Fruit cake. ZHU YUPEI. The invention discloses a fruit cake. The fruit cake comprises the following raw materials in parts by weight of 60-80 parts of wheat flour, 90-100 parts of eggs, 20-30 parts of mango granules, 40-50parts of cream, 20-30 parts of cherries, 10-15 parts of raisins, 20-30 parts of kiwi fruits and 20-30 parts of pineapples. The fruit cake disclosed by the invention is moderate in sour and sweet degrees, delicious, rich in nutrients and capable of beautifying the features and keeping young.

UA64910A "FRUIT" CAKE. KYIV NAT COMMERCIAL AND ECONOM [UA]. A cake contains extra wheat flour, granulated sugar, butter, melange, table salt, raisins, refined powder sugar, sodium bicarbonate. Additionally it contains a banana powder and water to recover the powder.

CN107736474A Dragon fruit cake. YANG YUXIANG. The invention provides a dragon fruit cake which is prepared from the following raw materials in parts by weight: 20-30 parts of dragon fruit jam, 20-30 parts of starch syrup, 5-8 parts of table salt, 3-4 parts of agar, 15-20 parts of honey, 0.8-1.0 part of citric acid, 0.1-0.5 part of malic acid and 0.1-0.2 part of sodium citrate. The taste of the dragon fruit cake prepared by the invention is between the tastes of jelly and soft sweets, and the dragon fruit cake is smooth and transparent in appearance, rich in elasticity, fine and smooth in taste, unique in taste, rich in nutrition and convenient to eat.

(11) Materials and Methods or Methodology (This includes the materials, methods, duration & location of the study. Materials and Methods are headings used in Natural Science researches and methodology is Social Sciences. Presentation of materials may not necessarily be in separate heading but may be incorporated in the specific methods used. Justify paragraphs with main headings in caps.)

The study used a score card as a research instrument. It looked into the four (4) sensory qualities such as appearance, aroma, taste and texture as the variables. In scoring the variables of the study, the Nine Point Hedonic Scale using a score card was used in evaluating the product.

Statistical Package for Social Sciences (SPSS) was used to generate and processed all the data needed in the study. Mean was used to determine the acceptability of the four sensory qualities tested of the products. The ANOVA was used in order to analyze the differences of the products in four sensory qualities among three treatments.

Three trials were done to come up with the most acceptable product in terms of sensory qualities such as appearance, aroma, taste, and texture. The finished product was evaluated with the semi-trained panelists for sensory evaluation and the acceptability was evaluated by the consumers.

The treatments used in the product formulation of miracle fruit cake were the following; Treatment A (20ml of miracle fruit wine), Treatment B (40ml of miracle fruit wine), and Treatment C

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015		Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH		Revision No.	00
				Effective Date	
				Page	Page 4 of 7

(60ml of miracle fruit wine).

The other ingredients for making cake were the following: all-purpose flour, baking soda, butter, brown and white sugar, eggs, fruits and raisins.

Product Formulation and Utilization

Procedure in the Preparation Miracle Fruit Wine. The fruit was selected, wash thoroughly, removing the pulp and weighed. Add wine yeast and muscovado. Cover primary fermentation bucket with cloth; allow must to ferment for one week to 10 days. Over the course of days, fermentation will cause a froth to develop on top and sediment to fall to the bottom. Gently strain the liquid to remove the sediment and froth. Run the juice through a funnel into sanitized glass secondary fermentation containers. Fill to the top to reduce the amount of air reaching the wine. Fit the containers with airlocks. Allow the juice to ferment for several weeks. Use the plastic tube to siphon the wine into clean glass secondary fermentation containers. Again, the purpose here is to separate the wine from sediment that forms as the wine ferments. Continue to siphon the wine off the sediment periodically (this is called "racking") for 2 or 3 months until the wine is running clear. Run the wine into bottles (using the cleaned plastic tubing), leaving space for the cork plus about a half inch or so of extra room. Insert corks. Store the wine upright for the first three days. After three days, store the wine on its side at, ideally, 55 degrees F

Procedure in the Preparation of Fruit and Raisin. Soaked the fruit and raisin in 120 ml of miracle fruit wine overnight. Then, it was drained and set aside for later use.

Procedure in the Preparation Miracle Fruit Cake. The ingredients were weighed. The all-purpose flour and baking powder was sifted. Then, the butter was creamed in the electric mixer until smooth, add the brown and white sugar little by little. While beating add the egg one at a time, then, add the dry ingredients while continuously beating until well-blended. Divide the batter mixture of cake into three parts. Each part of the batter mixture was added with miracle fruit wine, fruits, and raisins and it was combined until the mixture blended. The fruit cake mixture was placed in the lined-baking pan. The fruit cake mixture was baked in the preheated oven for 45 minutes or until done. The miracle fruit cake was removed from the oven and it was placed on the cooling rack to cool down. Then, placed the miracle fruit cake in a clean container and stored it overnight in the chiller.

(12) Results and Discussions (This part contains presentation of results (data) with reference to the table and figure, to be followed by discussion, with support of reviewed literature. Sequence shall be based on the objectives of the study. Tables are formatted without the vertical lines. Horizontal lines are set at the top and bottom portions as well as below table headings and above major subdivisions. Justify paragraphs with main headings in caps.)

Sensory Evaluation of Fruit Cake Infused with Miracle Fruit Wine

Table 1 shows the sensory evaluation of the 30 semi-trained panelists in terms of appearance, aroma, taste, texture and general acceptability. The data showed that in terms of appearance, aroma, taste and texture of the product Treatment C got a highest mean scores as described "Like Extremely" in all quality attributes. While Treatment B got a second highest mean scores, and Treatment C got the lowest mean scores among the three treatments.

This result implied that the higher the amount of miracle fruit wine infused in fruit cake was the most preferred by the evaluators since the flavor is delectable.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 5 of 7

Table 1. Sensory Evaluation of Fruit Cake infused with Miracle Fruit Wine.

QUALITY ATTRIBUTES	TREATMENTS					
	A		B		C	
	Mean	QD	Mean	QD	Mean	QD
Appearance	8.13	LE	8.40	LE	8.47	LE
Aroma	7.87	LVM	8.13	LE	8.33	LE
Taste	7.60	LVM	8.07	LVM	8.73	LE
Texture	7.67	LVM	8.07	LVM	8.60	LE
Gen.Acceptability	7.82	LVM	8.17	LE	8.53	LE

Legend:

Scale of Means

8.12 – 9.00

7.23 – 8.11

6.34 – 7.22

Qualitative Description

Liked Extremely

Liked Very Much

Liked Moderately

- LE

-LVM

-LM

Significant Difference in the Sensory Evaluation

Table 2 revealed the significant differences on the sensory evaluation of miracle fruit cake in terms of sensory qualities. The data revealed that there was no significant difference in the appearance in varying treatments as rated by the consumers ($F - \text{Value} = 1.620$, $P - \text{value} = .204 > .01$). Therefore, there was no difference exists in the appearance in the three treatments of the product, the three treatments was accepted. This implies that regardless of the treatments, the appearance of the products remained the same. As to its aroma, miracle fruit cake did not differed in their aroma ($F - \text{value} = 2.669$, $P - \text{value} = .075 > .01$). The researchers failed to reject the formulated hypothesis which indicate that there was no significant difference in aroma considering three treatments. This means that the treatments were similar in terms of having pleasant aroma. On the other hand, results revealed that there was a significant difference found in taste ($F - \text{value} = 24.240$, $P - \text{value} = .000 < 0.01$) and texture ($F - \text{value} = 8.449$, $P - \text{value} = .000 < 0.01$) among the treatments.

Therefore, the null hypothesis was rejected because the taste and texture of miracle fruit cake vary from one another and it was in favor of Treatment C. Furthermore, based on the general acceptability as a whole there was a significant difference among the three treatments of fruit cake infused with miracle fruit wine with the result shown ($F - \text{value} = 9.509$, $P - \text{value} = .000 < .01$) in favor of Treatment C.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 6 of 7

Table 2. Difference in the Consumer's Acceptability of the Miracle Fruit Cake.

SOURCES OF VARIANCE	SUM OF SQUARE	df	MEAN SQUARE	F-VALUE	P-VALUE	REMARKS
Appearance	1.867	2	.933	1.620	.204	ns.
	50.133	87	.576			
	52.000	89				
Aroma	3.289	2	1.644	2.669	.075	ns.
	53.600	87	.616			
	56.889	89				
Taste	19.467	2	9.733	24.240	.000	s.
	34.933	87	.402			
	54.400	89				
Texture	13.156	2	6.578	8.449	.000	s.
	67.733	87	.779			
	80.889	89				
Gen Acceptability	7.706	2	3.853	9.509	.000	s.
	35.250	87	.405			
	42.956	89				

P-value > .01 alpha

(13) Conclusions and Recommendations (This part shall be written in narrative form and arranged based on the objectives and findings. Conclusions are general statements answering the objectives. Recommendations are based on the conclusions. They may be split or combined based on their length, which means, conclusions and recommendations may be split under separate headings if they are long. Justify paragraphs with main headings in caps.)

Conclusions

Based on the findings of the study, the following conclusions are drawn:

Miracle fruit can be utilized in making fruit wine.

Miracle Fruit Cake is acceptable in terms of its sensory qualities.

Treatment C of Miracle Fruit Cake is the most preferred by the evaluators considering its sensory qualities.

There was a significant difference in the general acceptability of Miracle Fruit Cake in favor of Treatment C.

Recommendations

Based on the findings and conclusions of the study, the following recommendations are forwarded:

All treatments are recommended in making miracle fruit cake. Treatment C is the most preferred among the proportion.

Miracle fruit wine and the fruit cake may be subjected to microbial and proximate analysis, and its nutritional facts considering potential for product commercialization and technology utilization.

	Document Type:	DOCUMENTED INFORMATION ISO 9001:2015	Document Code	RES- CR-01
	Document Title:	COMPLETED RESEARCH	Revision No.	00
			Effective Date	
			Page	Page 7 of 7

Finally, it is recommended that further studies and applications of miracle fruit may be conducted in order to produce more varied food products aside from fruit cake for consumption and possible livelihood production.

(14) Output (Indicate the specific products, processes or services which the project has produced; quantify when possible.)

The output of this study is the production of Miracle Fruit Cake which is more healthy and less expensive compared to the existing commercial fruit cake.

(15) Technology Transfer Activity Conducted (Indicate the specific products, activity design, processes or services transferred as extension activities)

The product of this study could be one of the sources of extension activities conducted by the University, particularly on the main campus.

(16) Acknowledgments (Include only important persons and institutions with involvement in funding and conduct of the research. Justify paragraphs with main headings in caps.)

I would like to thank first the Lord our God who is always the source of my strength and wisdom in the pursuit of this research study. Likewise, I would like to thank all my co-authors, my fellow food technology professors, and evaluators for all the support they have extended throughout the conduct of this study. And also I would like to thank Capiz State University Key Officials, the Department of Industrial Technology, and the Research Office of the campus for the support extended for the success of this research endeavor. To God be the Glory.

(17) References (It should include only works mentioned in the text and should be arranged alphabetically. Citations shall not be less than 13 entries. This must follow the APA latest edition format, in hanging indentation with 12-point space in between.)

Calabash-the-miracle-fruit/<https://businessmirror.com/2015/10/15/>

Calabash-tree-tropical-fruits.php <https://www.fruitsinfo.com/>

Diseases-conditions/heart-disease/in-depth/red-wine/art-20048281: <https://www.mayoclinic.org/>

Miracle fruit: An alternative sugar substitute in sour beverages. Jéssica Ferreira Rodriguesa Rafaela da Silva Andradea Sabrina Carvalho Bastosb Sandra Bragança Coelhoa Ana Carla Marques Pinheiroa <https://www.sciencedirect.com/science/article/abs/pii/S0195666316304627>